

OS-MambaDiff: An Optical-SAR Fusion Diffusion Framework With Mamba for Robust Cloud Removal in Remote Sensing Imagery

Peizhen Yang^{1b}, Bo Huang^{1b}, *Member, IEEE*, Yuanxin Ye^{1b}, *Member, IEEE*, Jiacheng Zhang^{1b}, and Min Zhao

Abstract—Although optical remote sensing imagery has been extensively used in various Earth observation tasks, its practical utility is frequently compromised by cloud cover and adverse weather, which lead to missing data and disrupt the temporal continuity of observations. In contrast, Synthetic Aperture Radar (SAR) is an active remote sensing technology that functions independently of solar illumination and atmospheric conditions. Its ability to penetrate clouds and fog ensures the provision of reliable all-weather imagery. Motivated by the complementary characteristics of optical and SAR observations, we present OS-MambaDiff, a Mamba-based diffusion framework to fuse SAR information and restore cloud-contaminated optical images. This method can effectively fuse information from SAR and optical images without the need for a cloud mask, achieving high-precision cloud-free reconstruction. To this end, we design a Mamba-based Cross Modal Fusion Module, which introduces Mamba state space modeling and its 2-D extension SS2D to enable efficient feature fusion between heterogeneous modalities. Furthermore, a Cloud Loss based on differentiated constraint is designed. This loss function guides the model to learn cloud-free features while suppressing overfitting to cloud-contaminated condition images, thereby preventing the model from simply copying the conditional image output. In addition, a Sparsity-Aware Gated Fusion mechanism is constructed to adaptively inject fused SAR and optical features during the diffusion denoising process, while sparse regularization is employed to achieve stability and selectivity in feature modulation. Extensive experiments on the newly released cloud removal benchmark dataset SMILE-CR and the large-scale benchmark cloud removal dataset SEN12MS-CR show that OS-MambaDiff outperforms existing optical synthetic aperture radar cloud-removal techniques.

Index Terms—Cloud removal, deep learning, diffusion models, Mamba, optical-SAR fusion, remote sensing.

I. INTRODUCTION

IN RECENT years, high-resolution optical remote sensing technology has advanced rapidly, providing essential data support for applications such as land use monitoring and ecological environment assessment. However, its passive imaging mechanism relies on solar illumination and is highly susceptible to cloud occlusion and shadow interference. In particular, thick clouds severely attenuate surface reflectance and obscure spatial structure, fundamentally limiting the applicability of optical imagery in all-day and all-weather scenarios. In contrast, synthetic aperture radar (SAR) actively emits microwave signals [1], [2]. This study employs C-band Sentinel-1 SAR data (VV/VH polarization), which actively transmits microwave signals at a frequency of 5.405 GHz. This capability enables penetration through clouds and precipitation, providing stable structural information regardless of lighting conditions. Consequently, SAR serves as a natural complement to optical imagery. Nevertheless, SAR images exhibit strong speckle noise and lack direct correspondence to real-world optical spectral and textural characteristics, which significantly restricts their interpretability [66]. This makes the effective fusion of optical spectral richness and SAR structural reliability for high-fidelity cloud-free reconstruction a fundamental challenge in multimodal remote sensing.

Conventional physical modeling and image inpainting techniques struggle to capture the complex spectral–structural relationships under cloud cover. Although deep learning-based methods have improved single-modal cloud removal, they commonly rely on cloud masks or suffer from artifacts and texture degradation. To address these issues, we reformulate cloud removal as a conditional cloud-free image generation task and employ a diffusion model as the generative core. Due to its progressive denoising process, the diffusion model is particularly suited for this task, enabling stable optimization and fine texture recovery in complex remote sensing imagery. However, unlike general generation tasks, the conditional input (cloudy imagery) and target output (cloud-free imagery) in cloud removal exhibit high structural consistency in noncloudy regions. This strong correlation, driven by the model’s powerful reconstruction capabilities, often leads to an overreliance on conditional information, causing the model to neglect the inference of features beneath the clouds.

In light of this, directly applying diffusion models to cloud removal faces two major challenges: 1) How to efficiently and

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Peizhen Yang, Yuanxin Ye, and Jiacheng Zhang are with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu 611756, China (e-mail: ybacon999@gmail.com; yeyuanxin@home.swjtu.edu.cn; swjtu_zjc@163.com).

Bo Huang is with the Department of Geography, The University of Hong Kong, Hong Kong, SAR 999077, China, and also with the Smart City Research Institute, Shenzhen University, Shenzhen 518060, China (e-mail: bohuang@hku.hk).

Min Zhao is with the Department of Geography, The University of Hong Kong, Hong Kong, SAR 999077, China (e-mail: minzhao@hku.hk).

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adaptively fuse multimodal information to overcome the information deficiency of a single modality and 2) how to prevent the model from falling into a local optimum of “direct copying” of the cloudy input when conditional information (optical and SAR data) becomes overly dominant, which results in residual cloud artifacts.

To tackle these challenges, we propose OS-MambaDiff, a multimodal diffusion framework. Specifically, we design a Mamba-based Cross-Modal Fusion (MCMF) structure that leverages linear-complexity long-range modeling to achieve efficient optical-SAR feature interaction, making it highly suitable for large-scale remote sensing data processing. To address the degradation issue of identity copying, we further introduce a Cloud Loss based on differentiated constraint. This loss enforces structural differences between the generated output and the cloudy input at both the feature and pixel levels, compelling the model to focus on the substantive restoration of cloud-covered areas. Furthermore, we develop a Sparse-Aware Gated Fusion mechanism (SAGF), which adaptively identifies cloud-affected regions without the need for cloud masks and selectively integrates fused conditional information.

In summary, the main innovations and contributions of this work are as follows.

- 1) *Mamba-Based Cross-Modal Fusion Structure (MCMF)*: We are the first to introduce the Mamba dynamic modeling mechanism into Optical-SAR cloud removal. A two-dimensional state-space structure (SS2D), coupled with a Dynamic Fusion Feed-Forward Network (Dynamic Fusion FFN), enables efficient global dependency modeling and enhances SAR-guided cloud-free feature extraction.
- 2) *Cloud-Free Feature Loss (Cloud Loss) Based on Differentiated Constraint*: A differentiated constraint is imposed at pixel levels to mitigate conditional dominance, preventing the diffusion model from replicating cloudy inputs and guiding it to learn authentic cloud-free structures.
- 3) *Sparsity-Aware Gated Fusion (SAGF)*: The SAGF module automatically identifies cloud-affected regions and uses a sparsity constraint to drive gating values toward binary states. This mechanism enables the model to selectively integrate conditional features where necessary, thereby ensuring high accuracy and interpretability in multimodal feature fusion.

II. RELATED WORK

A. Cloud Removal

For decades, many researchers have focused on developing methods for removing clouds from optical remote sensing imagery. Traditional satellite image cloud removal techniques are mainly divided into three categories: Multispectral methods [5], [6], [7], multitemporal methods [8], [9], and interpolation-based methods [10], [11]. However, with the development of deep learning, the research focus has increasingly shifted toward neural network-based cloud removal techniques. For example, DSen2-CR [12] uses a residual network to establish a mapping relationship between cloud cover and cloud-free images, proposing a reconstruction model based on deep convolutional

neural networks. GLF-CR [2] introduces a global-local fusion method, which reduces noise in SAR images through dynamic filtering while embedding complementary information to assist cloud removal. SOSSF [13] employs multimodal data fusion techniques, proposing a dual-path fusion network based on attention mechanisms. This method integrates the spatial and spectral data from synthetic aperture radar and optical imagery to achieve cloud removal.

Furthermore, the introduction of Generative Adversarial Networks (GANs) [14] generates high-quality images by training a generator and a discriminator. Subsequent developments led to the emergence of Conditional Generative Adversarial Networks (cGANs) [15], which enable more controllable image generation. Recently, most cloud removal methods have relied on GAN technology. Some researchers use cGANs to directly translate cloud-free SAR images into optical images [16], [17] to achieve cloud removal. Spa-GAN [18] integrates a spatial attention mechanism within its GAN to improve the reconstruction quality in areas obscured by clouds. CTGAN [19] introduces Transformer-based GAN architecture to improve the quality of generated cloud-free images. ST-GAN [20] frames cloud removal as conditional image synthesis and utilizes a new spatiotemporal GAN architecture. MT-GAN [21] directly translates SAR images into optical images through multilayer transformation without the aid of optical image auxiliary information, thereby improving SAR interpretability and achieving cloud removal. SGFCR [63] proposes a SAR-assisted cloud removal network that exploits similarity-guided fusion and multiscale representation to transfer structural information from SAR images for accurate reconstruction of cloud-covered optical images. SADS-CR [64] introduce a supervised alignment perception and modality difference selection framework to alleviate spatial misalignment and modality-specific noise in SAR-optical fusion for cloud removal. GF-CR [65] develop a robust SAR-assisted cloud removal method that incorporates supervised align-guided fusion and bidirectional hybrid reconstruction to enhance structural consistency and detail recovery under complex cloud conditions.

However, limited by the inherent drawbacks of GAN adversarial training (e.g., difficulty in simultaneously optimizing generator and discriminator losses under dense cloud cover), training instability and mode collapse remain major challenges [22]. To overcome these bottlenecks, this study adopts a conditional diffusion model framework, which demonstrates a more stable training process and superior performance in the cloud removal task.

B. Diffusion Models

The diffusion process in diffusion models relies on a Markov chain [23], which iteratively introduces Gaussian noise until the image is completely transformed into random noise. The sampling process first learns the pattern of information degradation induced by noise. Based on this learned pattern, it then predicts the incrementally added noise to reconstruct the clean image. Diffusion models excel in high-resolution image generation [24], surpassing GANs [25] in synthesis quality and

diversity, establishing the paradigm of “forward noise addition and backward denoising”. This groundbreaking advancement has reshaped the landscape of cross-modal content generation, encompassing multiple domains including images [26], [27], [28], [29], text [30], [31], [32], and video [33], [34], [35]. Recently, many scholars have begun to explore cloud removal techniques based on the diffusion model paradigm. DDPM-CR [36] was the first to apply diffusion models to remote sensing image cloud removal, directly adopting the Denoising Diffusion Probabilistic Model (DDPM) [24]. DiffCR [37] combines conditional diffusion models with various optimization strategies to achieve high-quality cloud-free image generation. IDF-CR [38] uses a network based on Latent Diffusion Models (LDMs) for cloud removal, processing remote sensing images in the latent space via a Variational Autoencoder (VAE). DBD-CR [39] introduces a dual-branch diffusion residual network to achieve cloud removal. EDM-CR [40] adopts a unique enhanced diffusion model and designs a loss function suitable for the multiband nature of remote sensing images for cloud removal.

The above studies mainly rely on multitemporal optical images for cloud removal. Synthetic Aperture Radar (SAR) imagery, which can penetrate clouds, can provide key structural information about obscured surface features, offering important auxiliary information for generating cloud-free images. However, research on multimodal fusion methods that integrate optical and SAR imagery within the diffusion model framework is still relatively limited. DMDiff [41] proposes a dual-branch multimodal diffusion network, which guides cloud removal using fused Optical-SAR information. SAR-DeCR [42] combines diffusion models with a Transformer architecture to generate more precise cloud-free images based on SAR-derived surface information. RESTORE-DiT [43] is the first to propose a multitemporal optical-radar fusion diffusion model based on the Diffusion Transformer (DiT) network, reconstructing the real surface condition by comprehensively utilizing multitemporal information.

Inspired by these advancements, we integrate various optimization strategies, aiming to enhance cloud-free image generation quality through optical-radar fusion technology and improve the training efficiency of cloud removal diffusion networks.

III. METHOD

A. Problem Definition

The objective of the Optical-SAR fusion cloud removal task is to integrate ground surface information from one or more cloud-contaminated multispectral images and their corresponding SAR images over the same area, and ultimately produce cloud-free images of the region. The dataset is represented as $\mathbf{D} = \{(x_i, \text{Cloudy}_i, \text{SAR}_i)\}_{i=1}^N$. We denote $\mathbf{x}_i \in \mathbb{R}^{C_1 \times H \times W}$ as the cloud-free image. The dimensions C_1 , H , and W correspond to the number of spectral channels, height, and width of the image, respectively. Similarly, we define $\text{Cloudy}_i \in \mathbb{R}^{C_1 \times H \times W}$ as the cloud-contaminated image and $\text{SAR}_i \in \mathbb{R}^{C_2 \times H \times W}$ as the SAR image, both of which serve as conditional inputs to guide the generation of the cloud-free

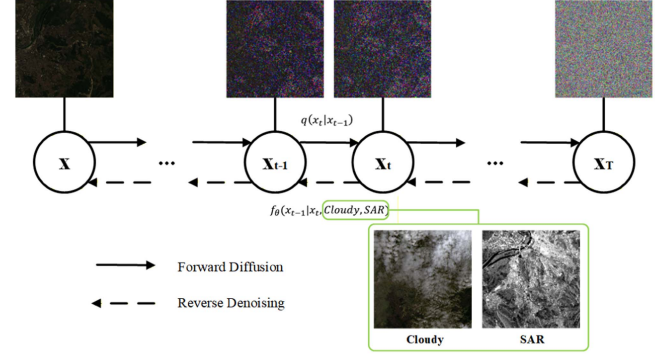


Fig. 1. Overall pipeline of our proposed cloud removal framework OS-MambaDiff.

image $\mathbf{x}_i$, with C_2 being the number of SAR image channels. Each triple $(\mathbf{x}_i, \text{Cloudy}_i, \text{SAR}_i)$ constitutes a matched set of images corresponding to the same geographical area, with precise pixel-level alignment among them. Furthermore, we denote the cloud-free image generated by our network as $\mathbf{y}_i \in \mathbb{R}^{C_3 \times H \times W}$, where C_3 represents the number of channels in the network’s output image.

B. Overall Framework

Inspired by DDPM [24], we propose an Optical-SAR fusion diffusion framework based on Mamba called OS-MambaDiff. As shown in Fig. 1, it consists of two processes: Forward Diffusion and Reverse Denoising. During model training, our framework takes cloud-contaminated images $\text{Cloudy} \in \mathbb{R}^{C_1 \times H \times W}$ and SAR images $\text{SAR} \in \mathbb{R}^{C_2 \times H \times W}$ as generation conditions, with a noisy cloud-free image $\mathbf{x} \in \mathbb{R}^{C_1 \times H \times W}$ at certain noise level t as input, the network is trained by minimizing the cloud-free feature loss based on differentiated constraints. After training, the framework generates high-quality cloud-free images of corresponding areas from randomly generated noise images and control conditions (such as cloud-contaminated images and SAR images).

1) *Forward Diffusion Process*: The forward diffusion process is a fixed, predefined Markov chain that gradually transforms an original cloud-free image $\mathbf{x}$ into an almost pure Gaussian noise image $\mathbf{x}_T$ within T steps, by adding a small amount of Gaussian noise at each step. This process requires no learning and is mathematically formulated as follows: At each step t , we generate a noisy image $\mathbf{x}_t$ according to a predefined variance schedule $\beta_t \in (0, 1)_{t=1}^T$, with the formula

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}). \quad (1)$$

This formula indicates that the current result is a scaled version of the previous step’s result, combined with Gaussian noise. A key design is that, thanks to the favorable properties of Gaussian distributions, we can directly compute the noisy image $\mathbf{x}_t$ at any timestep t from the original image $\mathbf{x}$ using the reparameterization trick, without requiring stepwise computation. The formula is

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x} + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad (2)$$

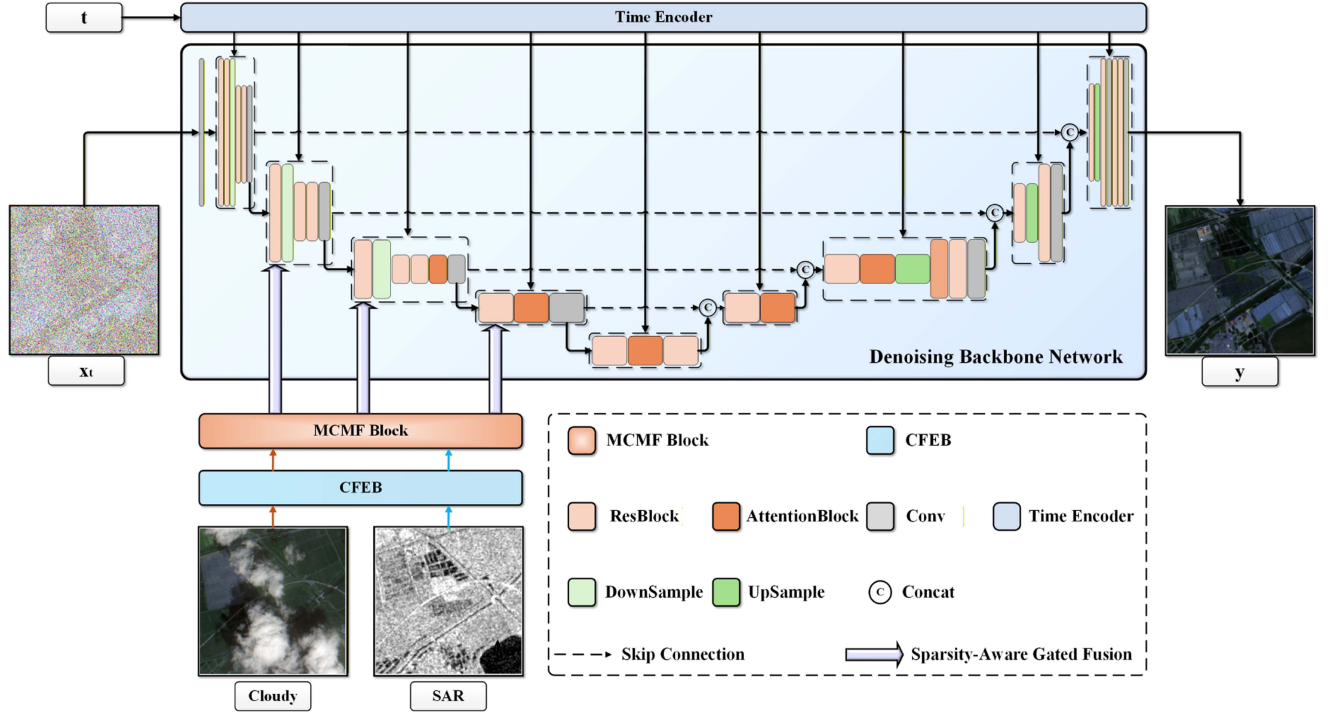


Fig. 2. Overall architecture of the conditional denoising model of OS-MambaDiff.

where $\bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$, and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is standard Gaussian noise. This shows that $\mathbf{x}_t$ is essentially a linear combination of the original image and random noise, with the signal-to-noise ratio controlled by $\bar{\alpha}_t$. As t increases, this ratio approaches zero, and eventually $\mathbf{x}_T$ approximates a standard Gaussian distribution.

2) *Reverse Denoising Process*: The goal of the reverse denoising process is to start from pure noise $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ and progressively remove noise to eventually generate a new sample $\mathbf{y}$ that conforms to the distribution of original cloud-free images. This is a Markov chain parameterized by a neural network, which needs to learn the reverse transition distribution at each step. Theoretically, when the noise β_t added at each step is sufficiently small, this reverse transition distribution also follows a Gaussian distribution. Therefore, most current denoising neural networks no longer aim to predict the image from the previous timestep but instead learn to predict the noise ϵ that was added to $\mathbf{x}_{t-1}$ to obtain $\mathbf{x}_t$ in the denoising process. The calculation formula is

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z} \quad (3)$$

where $\epsilon_{\theta}(\mathbf{x}_t, t)$ takes the current noisy image $\mathbf{x}_t$ and current timestep t as input and outputs its prediction of the added noise.

However, in the task of satellite remote sensing image cloud removal, predicting noise appears to be less efficient than directly predicting the data. Unlike general images, remote sensing images cover extensive areas and encompass complex geographic information data, resulting in complex image characteristics [60], [61]. Theoretically, directly predicting the cloud-free image $\mathbf{x}$ is equivalent to predicting the noise ϵ . However, in

satellite remote sensing image cloud removal tasks, our objective is to generate a cloud-free image that closely resembles the conditional image. Since the conditional image shares high similarity with the target image $\mathbf{x}$ to be predicted, selecting a direct data prediction method proves more convenient and converges faster. If the neural network can effectively predict the data $\mathbf{x} = f_{\theta}(\mathbf{x}_t, \text{Cloudy}, \text{SAR}, t)$, then formula (2) can be inverted to derive the estimated noise value ϵ

$$\epsilon = \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}}{\sqrt{1 - \bar{\alpha}_t}}. \quad (4)$$

Substituting into formula (3), it transforms into

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \left(\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}}{\sqrt{1 - \bar{\alpha}_t}} \right) + \sigma_t \mathbf{z}. \quad (5)$$

Now, the learning objective of the model shifts from “how to progressively denoise” to “how to directly reconstruct clean data from data at any noise level.”

C. Conditional Denoising Model

The proposed OS-MambaDiff (as shown in Fig. 2) is a multimodal generative network based on the diffusion process, which achieves high-fidelity reconstruction of cloud-free remote sensing images by integrating optical and SAR information. The entire network mainly consists of four components: 1) Conditional Feature Extraction Block (CFEB) for feature encoding of optical and SAR modalities; 2) Mamba Cross-Modal Fusion Module (MCMF) for multimodal feature fusion; 3) Time Encoder for adding random noise; and 4) OS-MambaDiff Denoising Backbone Network for diffusion-based cloud-free image generation.

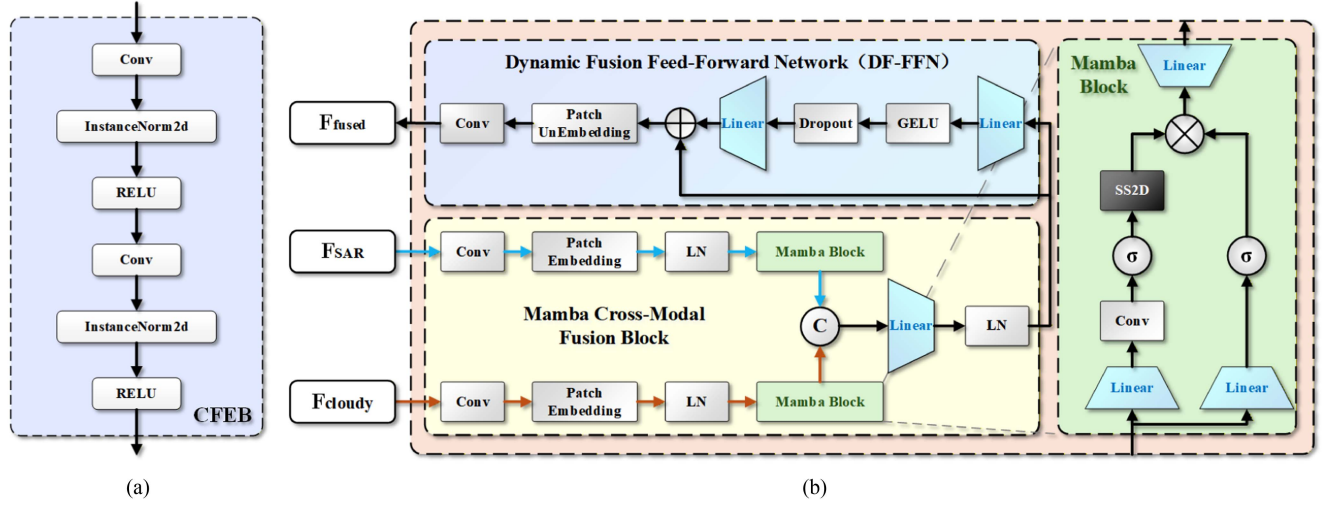


Fig. 3. (a) Schematic of our proposed CFEB. (b) Schematic of our proposed MCMF block.

1) *Conditional Feature Extraction Block*: The CFEB module [as shown in Fig. 3(a)] aims to extract multiscale features from both optical cloudy images and SAR images. Given a cloudy optical image **Cloudy** and the corresponding SAR image **SAR**, CFEB extracts their respective spectral and spatial features as follows:

$$\mathbf{F}_{\text{SAR}} = \text{CFEB}(\text{SAR}), \mathbf{F}_{\text{Cloudy}} = \text{CFEB}(\text{Cloudy}). \quad (6)$$

In this process, the optical branch preserves spectral semantic information, while the SAR branch captures structural texture and surface scattering features. In this way, the model captures optical texture details and SAR structural information at different spatial scales, laying a solid foundation for subsequent fusion.

2) *Mamba Cross-Modal Fusion Module (MCMF)*: We adopt Mamba [44] as the basic framework for multimodal Optical-SAR fusion. The Mamba model introduces a selective state-space model, achieving strong modeling capacity while maintaining linear complexity. Mamba has been widely applied in the field of remote sensing [44], [45], [46], [47], [48], [49], [50] and has shown remarkable potential. To efficiently fuse structural information from the **SAR** image with cloudy contextual information from the **Cloudy** image, we design a multimodal feature fusion module MCMF [as shown in Fig. 3(b)], inspired by Vision Mamba [51], FusionMamba [52], and STFMamba [53]. The module consists of a Mamba Cross-Modal Fusion Block and a Dynamic Fusion Feed-Forward Network.

a) *Mamba Cross-Modal Fusion Block*: This subblock is responsible for serial modeling and initial fusion of multimodal features $\mathbf{F}_{\text{SAR}}$ and $\mathbf{F}_{\text{Cloudy}}$. First, the features of both modalities are preprocessed as follows:

$$\mathbf{F}'_{\text{SAR}}, \mathbf{F}'_{\text{Cloudy}} = \text{LN}(\text{PatchEmbedding}(\text{Conv}(\mathbf{F}_{\text{SAR}}, \mathbf{F}_{\text{Cloudy}}))). \quad (7)$$

Then, the preprocessed features are fed into their respective Mamba Blocks. The Mamba Block is based on the Selective Scan in Two Dimensions (SS2D) structure to better adapt to the spatial properties of image data. SS2D extends the state space

model to two-dimensional spatial data by modeling long-range dependencies along both spatial dimensions through selective scanning, enabling efficient global context modeling for images. The SS2D structure enables Mamba to capture global dependencies in feature maps with linear complexity while preserving rich long-range contextual information. After Mamba modeling, the two modal features $\mathbf{M}_{\text{SAR}}$ and $\mathbf{M}_{\text{Cloudy}}$ are concatenated, followed by a projection and normalization layer to obtain the fused sequence feature $\mathbf{F}_{\text{seq}}$, which is then input into the DF-FFN.

b) *Dynamic Fusion Feed-Forward Network (DF-FFN)*: To further enhance and modulate the fused features, we introduce DF-FFN, a standard Feed-Forward Network (FFN) structure with residual connections to strengthen feature representation. The computation is defined as

$$\mathbf{F}_{\text{FFN}} = \text{Linear}(\text{Dropout}(\text{GELU}(\text{Linear}(\mathbf{F}_{\text{seq}})))) + \mathbf{F}_{\text{seq}}. \quad (8)$$

Finally, $\mathbf{F}_{\text{FFN}}$ passes through Patch UnEmbedding and Conv layers to map the sequence features back into the image feature space, obtaining the final fused feature $\mathbf{F}_{\text{fused}}$. $\mathbf{F}_{\text{fused}}$ serves as a key cloud-free feature guidance signal, laterally injected into different levels of the backbone network to guide denoising (cloud removal) during the diffusion process.

3) *Time Encoder*: Random-step noise is added to the cloud-free image as training input to enhance data unpredictability and improve network reasoning capability. Therefore, the noise step t is used for auxiliary training and incorporated into the model via the time encoder. Following the approach in [24], [25], [27], and [54], the time encoder is defined as

$$\mathbf{F}_t = \text{Reshape}(\text{MLP}(\sin(t))) \quad (9)$$

where t denotes the time step, $\sin(\cdot)$ represents sinusoidal encoding, and MLP is a multilayer perceptron mechanism. Reshape() adjusts the time embedding to match the image feature dimensions, and $\mathbf{F}_t$ denotes the time embedding.

4) *OS-MambaDiff Denoising Backbone*: The denoising backbone of our OS-MambaDiff model is built on a standard U-Net architecture within a conditional diffusion framework.

In each diffusion step, the network receives a noisy cloud-free image $\mathbf{x}$, the time embedding $\mathbf{F}_t$, and multimodal fused features $\mathbf{F}_{\text{fused}}$ from MCMF. The fused information is injected at corresponding scales, and the network outputs the predicted cloud-free image $\mathbf{y}$. We choose to inject MCMF fusion features after each DownSample operation rather than at the very beginning. This strategy prevents the model from accessing conditional information too early, thereby reducing dependence on the conditional image and preventing shortcut learning through direct copying.

a) U-Net Encoder–Decoder Structure: The backbone network adopts a symmetric encoder–decoder structure that progressively extracts and reconstructs features through a series of DownSample and UpSample operations. The key processing components are as follows.

- 1) *ResBlock:* We employ a time-conditioned residual block. The input feature first passes through Norm, SiLU, and Conv layers for preprocessing and is then added with the time embedding $\mathbf{F}_t$. This design ensures that the network can accurately modulate feature representations based on the current diffusion time step.
- 2) *AttentionBlock:* At the intermediate stage (typically the lowest feature resolution), we use an AttentionBlock to capture global contextual information and spatial dependencies within the feature maps, facilitating the reconstruction of complete surface structures under large-area cloud occlusion.

b) Sparsity-Aware Gated Fusion (SAGF): Multimodal fused features $\mathbf{F}_{\text{fused}}$ are injected into the encoder part of the backbone network through our proposed Sparsity-Aware Gated Fusion (SAGF) mechanism. Specifically, SAGF uses the cross-modal fusion feature $\mathbf{F}_{\text{fused}}$ provided by MCMF to generate an adaptive gating weight gate, which determines the proportion of fusion between $\mathbf{x}$ and the original feature $\mathbf{F}_{\text{fused}}$

$$\text{gate} = \text{Sigmoid}(\text{Conv}(\text{Concat}(\mathbf{x}, \mathbf{F}_{\text{fused}}))) \quad (10)$$

$$\mathbf{x}' = \mathbf{x} \cdot (1 - \text{gate}) + \mathbf{F}_{\text{fused}} \cdot \text{gate}. \quad (11)$$

Here, $\mathbf{x}$ represents the output feature from the previous decoder module, and $\mathbf{F}_{\text{fused}}$ is the cloud-free guidance feature laterally provided by MCMF. Combined with the sparsity regularization loss $L_{\text{gate_sparse}}$, the gate is encouraged to produce sparse, near-binary activations, making gating decisions more explicit. This effectively suppresses the propagation of cloud-contaminated information and ensures high-quality feature fusion.

D. Training Objectives and Loss Functions

1) Cloud-Free Feature Loss (Cloud Loss) Based on Differentiated Constraint: In multimodal cloud removal tasks, cloudy optical images typically exhibit strong spatial correlations with the target images. Without explicit supervision (e.g., cloud masks), the model may degenerate into a trivial mapping $\mathbf{y} \approx \text{Cloudy}$, leading to distorted or nonreconstructed results. To mitigate this, we construct a contrastive constraint mechanism that enhances differentiated learning in cloud-covered areas while maintaining reconstruction consistency in cloud-free regions,

guiding the model to reconstruct realistic and distinct cloud-free features.

First, we define the cloud region indicator

$$\mathbf{I}_{\text{cloud}}(i) = \text{mean}(|\text{Cloudy}_i - \mathbf{x}_i|, \text{dim} = 1, \text{keepdim} = 1) \quad (12)$$

which measures the degree of cloud contamination per pixel.

Then, we compute the difference between the generated image and the conditional image

$$\mathbf{D}_{\text{diff}}(i) = |\mathbf{y}_i - \text{Cloudy}_i|. \quad (13)$$

The Cloud Loss is defined as

$$L_{\text{cloud}} = \text{mean}(\mathbf{I}_{\text{cloud}}(i) \cdot \lambda_{\text{cloud}} \cdot \text{ReLU}(m - \mathbf{D}_{\text{diff}}(i))) \quad (14)$$

where λ_{cloud} is the cloud contamination weight, and m is the contrast margin (we use $m = 0.2$). When the generated image and the cloudy image are too similar in cloud regions (i.e., $\mathbf{D}_{\text{diff}}(i) < m$), the penalty is activated, forcing the model to enhance differentiated reconstruction in these regions. In cloud-free areas, since $\mathbf{I}_{\text{cloud}}(i)$ is small, the loss is naturally suppressed, maintaining spectral consistency. The proposed method is trained under a paired supervised setting, where cloudy optical images and their corresponding cloud-free ground truth images are available. The cloud region indicator used in the Cloud Loss is computed from ground-truth data and serves as a soft spatial weighting mechanism during training only. During inference, neither ground-truth information nor cloud masks are required.

2) Sparsity-Aware Gating Loss: This loss specifically regularizes the gating maps used in the SAGF mechanism, encouraging them to approach binary extremes (0 or 1), thereby enforcing more decisive feature selection. For the i th sample, the gating output sequence is $\{\text{gate}_k\}_{k=1}^K$ (K represents gates at different layers), and the sparsity regularization loss is defined as

$$L_{\text{gate_sparse}}(i) = \frac{1}{K} \sum_{k=1}^K \text{mean}(\text{gate}_k(i) \cdot (1 - \text{gate}_k(i))). \quad (15)$$

The overall gating loss averaged across samples is

$$L_{\text{gate_sparse}} = \frac{1}{N} \sum_{i=1}^N L_{\text{gate_sparse}}(i). \quad (16)$$

3) Total Loss: To ensure pixel-level reconstruction accuracy, the basic reconstruction objectives consist of MSE and SSIM losses. The mean squared error between $f_{\theta}(\mathbf{x}_t, \text{Cloudy}, t)$ and the ground truth cloud-free image $\mathbf{x}$ is

$$L_{\text{MSE}} = \mathbb{E}_{\mathbf{x}, \epsilon} \|\mathbf{x} - f_{\theta}(\sqrt{\alpha_t}\mathbf{x} + \sqrt{1 - \alpha_t}\epsilon, \text{Cloudy}, t)\|^2 \quad (17)$$

and the structural loss is

$$L_{\text{SSIM}} = 1 - \text{SSIM}(\mathbf{x}, f_{\theta}(\mathbf{x}_t, \text{Cloudy}, t)). \quad (18)$$

Thus, the total loss function is

$$L_{\text{total}} = \lambda_{\text{MSE}} \cdot L_{\text{MSE}} + \lambda_{\text{SSIM}} \cdot L_{\text{SSIM}} + \lambda_{\text{gate}} \cdot L_{\text{gate_sparse}} + \lambda_{\text{cloud}} \cdot L_{\text{cloud}} \quad (19)$$

where λ_{MSE} and λ_{SSIM} are pixel and structural reconstruction weights ($\lambda_{\text{MSE}} + \lambda_{\text{SSIM}} = 1$, we use $\lambda_{\text{MSE}} = 0.5$ and $\lambda_{\text{SSIM}} =$

0.5). λ_{gate} is the sparsity-aware gating loss weight ($\lambda_{\text{gate}} \in [0.005, 0.05]$, set to $\lambda_{\text{gate}} = 0.005$). λ_{cloud} is the cloud-free differentiated constraint weight ($\lambda_{\text{cloud}} \in [0.3, 0.7]$, set to $\lambda_{\text{cloud}} = 0.3$). Constrained by the vast coverage and complex image characteristics of remote sensing imagery, we employ a data prediction approach. However, this method simultaneously introduces a conditional collapse issue, which we mitigate by treating L_{cloud} as a differentiable constraint.

IV. EXPERIMENTS

A. Datasets

We used the newly released high-resolution cloud removal dataset SMILE-CR [3] and the large-scale general cloud removal dataset SEN12MS-CR [4] to validate the effectiveness of the network's multimodal fusion for cloud removal.

1) *SMILE-CR*: This study employed the recently released SMILE-CR cloud removal benchmark dataset for model training and evaluation. SMILE-CR is a globally distributed multimodal dataset aimed at solving cloud removal for Landsat-8 optical imagery. The dataset was created using a realistic cloud simulation technique. It comprises Landsat-8 optical pairs (cloudy and cloud-free), Sentinel-1 SAR imagery, and auxiliary MODIS multispectral information, along with precise cloud masks. We use 1000 samples for training, 200 for validation, and 200 for testing, following the division suggested by the official dataset. Each pair of images is $512 \times 512 \text{ px}^2$ in size. The dataset images are saved in original TIFF format and possess high image resolution and quality.

2) *SEN12MS-CR*: This dataset is an upgrade of SEN12MS [55], being the first cloud-free dataset with global range and all-season coverage. It contains paired Sentinel-2 multispectral optical satellite observations that are nearly cloud-free and cloud-covered, along with corrected Sentinel-1 spaceborne radar measurements. Cloud-free Sentinel-2 images were selected with a cloud cover rate below 10%. Cloudy images have a cloud cover range between 20% and 70%. Optical images are all from Level-1 C Top-of-Atmosphere reflectance products, having 13 bands. SAR images are from Level-1 GRD products, having two polarization channels: VV and VH. SEN12MS-CR contains observations from 175 globally distributed areas of interest recorded in one season across the four seasons in 2018. The full scene images are divided into 122 218 triplets. Each pair of images is the same size of $256 \times 256 \text{ px}^2$, and the pixels are aligned. These three images were acquired within the same meteorological season to minimize changes in surface features. The data captures a rich spectrum of cloud conditions, ranging from clear scenes used for verification and training, semitransparent fog, and extensive dense cloud cover.

B. Implementation Details and Experimental Results

For the experimental environment and implementation details, all experiments were completed on a high-performance computing system equipped with four NVIDIA GeForce RTX 4090 graphics cards (single card 24 GB memory), dual AMD EPYC 7543 processors, and 480 GB memory, running on Ubuntu 22.04 LTS. The model was implemented based on

the PyTorch 2.1.0 framework (CUDA 12.1, Python 3.10) and trained on all four 4090 GPUs using Distributed Data Parallel (DDP). During training, we used the AdamW optimizer (weight decay 0), with an initial learning rate of 5×10^{-5} , a minimum learning rate of 1×10^{-8} , and a Sigmoid [56] strategy for learning rate scheduling. The total batch size was set to 20. The model was trained for 1000 epochs, totaling 1M steps. In the diffusion stage, we used Sigmoid schedules to increase noise and set the maximum noise step to 2000. We also used exponential moving average (EMA) [57] with a decay rate of 0.999 to prevent overfitting and increase training stability. For the SMILE-CR dataset setup, we downsampled the SMILE-CR data to $256 \times 256 \text{ px}^2$ and normalized it to the range $[0, 1]$. It should be noted that the cloud mask provided by the SMILE-CR data set is used exclusively for evaluation and regional metric analysis. It is not utilized in model training, loss calculation, feature fusion, gating processing, or inference.

Our evaluation of the cloud removal models relied on a suite of metrics, including the Structural Similarity Index Measure (SSIM), Peak Signal-to-Noise Ratio (PSNR), Frechet Inception Distance (FID), and LPIPS, to gauge performance quantitatively. We compared Pix2Pix [62], DiffCR [37], DSen2-CR [12], DDPM-CR [36], GLF-CR [2], IDF-CR [38], DBD-CR [39], McGAN [58], UnCRtainTS [59]. To ensure fair comparison, all methods evaluated in this study—including the proposed OS-MambaDiff and all baseline approaches—were trained and tested at a uniform spatial resolution of 256×256 . Furthermore, all experiments were conducted in identical computational environments using identical hardware resources and distributed training configurations. The baseline model is implemented or retrained according to official specifications, with input resolution adjusted to 256×256 to ensure consistency. According to the original dataset documentation [3], [4], both optical images and synthetic aperture radar images underwent terrain correction and spatial alignment prior to region extraction. Table I shows the performance metrics of our method and other methods on the SMILE-CR and SEN12MS-CR datasets. It can be seen in Table I that OS-MambaDiff outperforms other methods in all metrics, with PSNR = 28.9243, SSIM = 0.9026, FID = 51.4184, and LPIPS = 0.1119 on the test set of the SMILE-CR dataset, and PSNR = 30.5585, SSIM = 0.8951, FID = 91.5629, and LPIPS = 0.2399 on the test set of SEN12MS-CR (as shown in Table II). Fig. 4 compares the cloud removal results generated by our method with other methods. Since the SMILE-CR dataset contains some cloud occlusion-free samples, we selected several samples from the test set for visualization. These samples all contain a large amount of thick cloud cover, making the cloud removal task difficult. OS-MambaDiff still achieves excellent cloud removal effects and image generation quality under the harsh challenge of thick cloud occlusion. It outperforms other methods in terms of structure generation in cloud-covered areas and overall image brightness recovery, achieving the highest image fidelity.

C. Ablation Study

This section empirically analyzed the contribution of each key component in OS-MambaDiff through controlled ablation

TABLE I
QUANTITATIVE COMPARISON OF CLOUD REMOVAL PERFORMANCE AND EFFICIENCY BETWEEN OS-MAMBADIFF AND EXISTING MODELS BASED ON THE SMILE-CR DATASET

Method	PSNR $\uparrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$	Param (M)	MACs (G)
DSen2-CR [12]	25.5008	0.7456	189.6285	0.4298	18.93	620.41
DDPM-CR [36]	25.1284	0.7213	161.5738	0.4855	445.44	426.18
DiffCR [37]	26.7029	0.7764	140.6443	0.3734	22.91	22.93
GLF-CR [2]	27.5124	0.8092	120.7888	0.3521	14.83	981.11
IDF-CR [38]	28.7865	0.8282	112.9872	0.3126	73.27	207.36
DBD-CR [39]	28.4812	0.8892	98.1895	0.2978	16.76	18.85
OS-MambaDiff (Ours)	28.9243	0.9026	51.4184	0.1119	81.70	125.08

The bold values indicate the performance of the method.

TABLE II
QUANTITATIVE COMPARISON OF CLOUD REMOVAL PERFORMANCE AND EFFICIENCY BETWEEN OS-MAMBADIFF AND EXISTING MODELS BASED ON THE SEN12MS-CR DATASET

Method	PSNR $\uparrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$	Param (M)	MACs (G)
McGAN [58]	24.8025	0.7421	178.2912	0.4295	54.40	142.27
DSen2-CR [12]	27.2875	0.8721	160.2166	0.3921	18.93	620.41
DDPM-CR [36]	28.9001	0.8734	164.6245	0.4123	445.44	426.18
DiffCR [37]	29.5211	0.8802	132.1225	0.3209	22.91	22.93
GLF-CR [2]	28.9021	0.8830	104.2452	0.2820	14.83	122.64
SAR-DeCR [42]	27.9240	0.8459	159.9286	0.3922	852.03	4421.51
RESTORE-DiT [43]	29.2881	0.8821	98.3976	0.2581	10.09	196.63
UnCRtainTS [59]	28.9599	0.8945	121.9559	0.2993	0.56	37.16
OS-MambaDiff (Ours)	30.5585	0.8951	91.5629	0.2399	81.70	125.08

The bold values indicate the performance of the method.

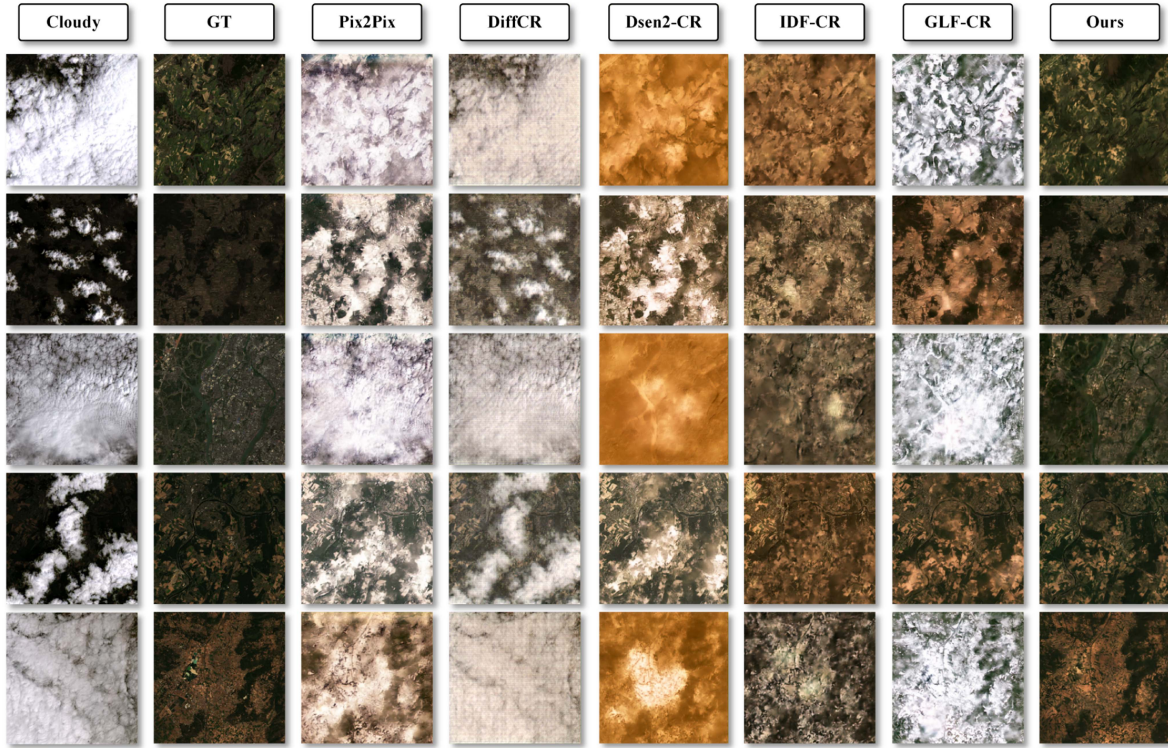


Fig. 4. On the SMILE-CR dataset, we qualitatively compare the visual results of our proposed OS-MambaDiff with other existing methods. Here, GT denotes ground truth for cloud-free imagery.

experiments on the SMILE-CR dataset. The objective is to quantify the extent to which each design choice contributes to performance, while explanations of the underlying mechanisms will be discussed separately in Section IV-D.

1) *Mamba Cross-Modal Fusion Block*: To evaluate the effectiveness of multimodal fusion, we compare several fusion

strategies under identical diffusion backbones and training settings. These include an optical-only baseline, simple convolutional concatenation, Transformer-based fusion, and the proposed Mamba Cross-Modal Fusion (MCMF).

Results in Table III indicated that excluding SAR information leads to a substantial performance drop, confirming the

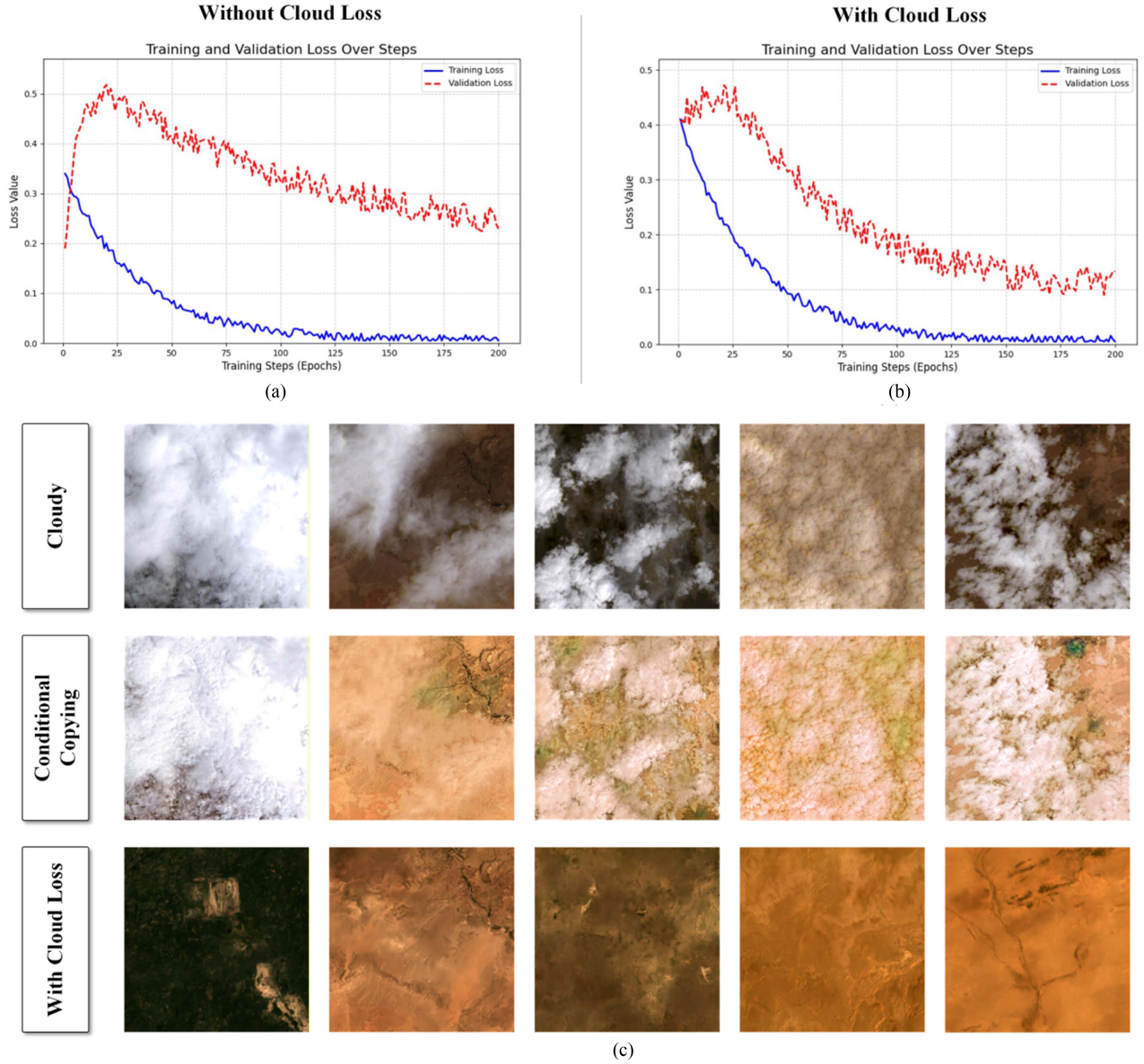


Fig. 5. Fig. (b) shows the model loss over training epochs using Cloud-Free Feature Loss based on Differentiated Constraint, while (a) shows the same without it. The solid blue line indicates the training set loss, and the dashed red line indicates the validation set loss. Fig. (c) shows the conditional replication phenomenon in data-predictive diffusion models and the suppression effect of Cloud Loss on degenerative behavior.

TABLE III
EVALUATION SCORES FOR DIFFERENT GATE LAYERS BASED ON CLOUD MASK DATA FROM THE SMILE-CR DATASET

Fusion Model	PSNR $\uparrow$	SSIM $\uparrow$	Param (M)	MACs (G)	Runtime (ms)
Optic Only	18.2341	0.5212	37.91	49.77	7.8
Conv Concat	22.6455	0.6409	55.60	92.01	12.2
Transformer	28.0199	0.8342	114.42	415.96	103.2
MCMF Block	28.9243	0.9026	81.70	125.08	21.5

necessity of structural guidance under dense cloud conditions. While convolutional concatenation introduces SAR cues, it fails to fully exploit long-range dependencies. Transformer-based fusion improves reconstruction quality but incurs significantly higher computational cost. In contrast, MCMF achieves superior performance with a more favorable balance between accuracy, parameter efficiency, and runtime.

2) *Cloud-Free Feature Loss Based on Differentiated Constraint*: This experiment evaluates the influence of the proposed cloud-free feature loss on training behavior and reconstruction quality. Models trained without this loss exhibit a strong tendency toward conditional replication during early training stages, resulting in unstable optimization and degraded cloud reconstruction.

TABLE IV
SENSITIVITY ANALYSIS OF PARAMETERS m AND λ_{cloud} IN CLOUD LOSS
BASED ON THE SMILE-CE DATASET

	m	λ_{cloud}	PSNR $\uparrow$	SSIM $\uparrow$
baseline	0	0	26.4215	0.8012
groupA	0.1	0.3	27.4152	0.8446
	0.2	0.3	28.9243	0.9026
	0.3	0.3	28.1056	0.8731
groupB	0.2	0.1	28.2035	0.8718
	0.2	0.5	28.5239	0.8832
	0.2	0.7	27.869	0.8627

The bold values indicate the performance of the method.

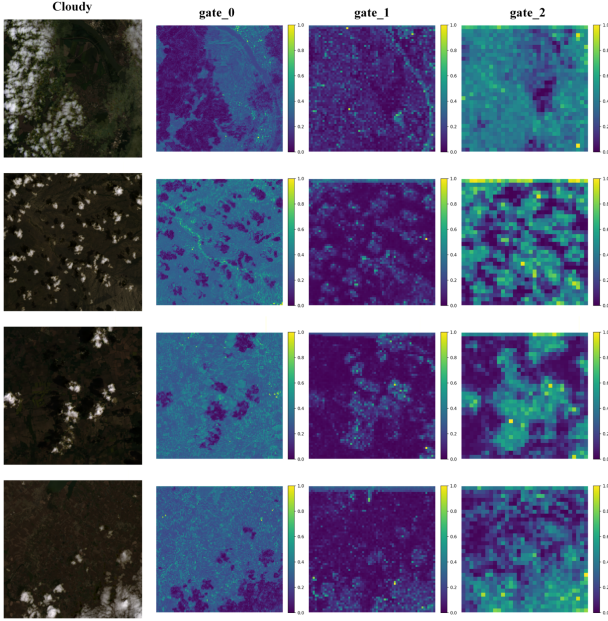


Fig. 6. Visual comparison of gating results, where gate_0 represents the shallow-level mapping, gate_1 represents the deep-level mapping, and gate_2 represents the deeper-level mapping.

As shown in Fig. 5, introducing the differentiated constraint leads to smoother loss trajectories and improved convergence stability. Sensitivity analysis in Table IV further shows that the margin parameter plays a dominant role in enforcing effective differentiation, while the loss weight primarily regulates training intensity. These results demonstrate that the proposed loss provides a practical mechanism for mitigating conditional collapse in data-predictive diffusion models.

3) *Sparsity-Aware Gated Fusion Mechanism*: To evaluate the effectiveness of SAGF, we simultaneously analyzed the consistency between the qualitative gating pattern and the quantitative true cloud mask. For visualization purposes, the gating activation values across all layers underwent joint normalization to preserve relative amplitude differences between layers. This gating mechanism effectively suppresses unnecessary condition fusion in clear regions while highlighting cloud-affected areas.

As shown in Fig. 6 and Table V, quantitative evaluation confirms that gating predictions achieve high accuracy across multiple network depths. Moreover, different layers exhibit distinct gating behaviors, reflecting scale-dependent fusion strategies. These findings validate that SAGF enables selective and spatially adaptive feature integration without explicit cloud supervision.



Fig. 7. Loss trends over training steps for data prediction (solid green line) and noise prediction (dashed yellow line).

4) *Cloud Region Indicator of Cloud Loss*: To further elucidate the role of cloud region indicator in the proposed cloud loss model and assess its dependence on ground-based observational data, it is necessary to examine the model's performance in the absence of ground-truth information. We compare three variants using identical training settings on the SMILE-CR dataset. The full model (GT indicator) employs the original formula. Uniform weighting (without cloud-aware weighting) replaces the cloud region indicator with a constant value of 1 for all pixels, effectively eliminating spatial variability and simplifying cloud loss to a global boundary constraint. Spectral weights (GT-independent) employ a brightness-based thresholding strategy, computing a simple cloud estimate directly from the cloud optical image.

The results in Table VI indicate that replacing GT-based cloud region indicator leads to a moderate performance decline. However, GT-free alternatives remain viable, demonstrating that the Cloud Loss framework is fundamentally independent of actual cloud maps. Cloud region indicator effectively function as training stabilizers and optimization enhancers.

5) *Advantages of Data Prediction*: Finally, we compare data prediction and noise prediction strategies within the diffusion framework. Empirical results in Fig. 7 show that directly predicting cloud-free images leads to faster convergence and more stable optimization than predicting noise. This advantage is particularly pronounced in conditional cloud removal tasks, where large portions of the target image are already observable in the conditional input.

D. Discussion

This section discusses the proposed OS-MambaDiff framework from a conceptual and methodological perspective, with an emphasis on understanding why the proposed design choices are effective for Optical-SAR cloud removal, rather than reiterating quantitative comparisons or ablation results presented in Section IV-C.

1) *Robustness Across Diverse Cloud Conditions*: Cloud removal performance can vary significantly across different cloud morphologies and densities, which poses a critical challenge for practical deployment. To contextualize the robustness of OS-MambaDiff under such heterogeneous conditions, Fig. 8 presents representative qualitative results in a wide range of

TABLE V
QUANTITATIVE EVALUATION OF CLASSIFICATION ACCURACY USING THE CLOUD MASK (EVALUATION ONLY) WITH SMILE-CR

Gate Layer	OA(Overall Accuracy) $\uparrow$	F1(F1-score) $\uparrow$	PA(Recall) $\uparrow$	UA(Precision) $\uparrow$
gate_0	87.73%	0.8908	86.59%	91.72%
gate_1	83.23%	0.8510	82.16%	88.25%
gate_2	80.99%	0.8133	78.94%	83.87%
Derivation	$TP + TN$	$2 \times \text{Precision} \times \text{Recall}$	TP	TP
	$TP + TN + FP + FN$	Precision + Recall	$TP + FN$	$TP + FP$

TABLE VI
IMPACT OF CLOUD REGION INDICATOR ON MODELS IN THE ABSENCE OF GROUND TRUTH DATA

Variant	PSNR $\uparrow$	SSIM $\uparrow$
Full Model	28.9243	0.9026
Uniform Weighting	27.9880	0.8746
Spectral Weights	28.4522	0.8893

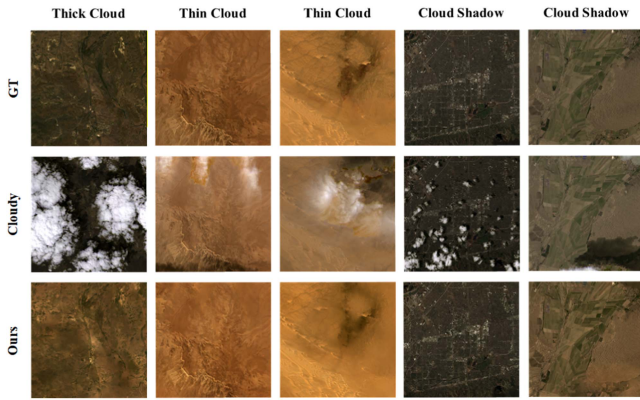


Fig. 8. Qualitative comparison of cloud removal performance under varying cloud conditions. This figure summarizes the model's performance across diverse scenarios, encompassing different cloud density levels.

cloud scenarios, including thin clouds, translucent clouds, dense clouds, and patchy cloud patterns.

Rather than serving an additional experimental validation, these examples illustrate the stability of the proposed framework when confronted with varying degrees of information loss and structural ambiguity. The consistently coherent reconstructions across all scenarios suggest that OS-MambaDiff effectively balances SAR-derived structural guidance and optical spectral refinement, enabling reliable restoration under diverse cloud conditions.

2) *Optical-SAR Fusion Under the Diffusion Paradigm*: The strong performance of OS-MambaDiff highlights the intrinsic advantage of combining Optical and SAR modalities within a diffusion-based generative framework. In cloud removal tasks, optical imagery preserves spectral semantics in cloud-free regions, while SAR provides robust geometric and structural cues beneath clouds. Diffusion models are particularly well-suited to exploit this complementarity, as their iterative denoising process enables gradual refinement from coarse structure to fine texture. This characteristic allows SAR-derived structural priors to guide early-stage reconstruction, while optical information progressively restores spectral fidelity.

3) *Rationale for Mamba-Based Cross-Modal Fusion*: The adoption of Mamba for cross-modal feature fusion is motivated by the need for efficient long-range dependency modeling in large-scale remote sensing imagery. Compared with attention-based Transformers, Mamba's selective state-space formulation achieves linear complexity while retaining global context modeling capability. This makes it particularly suitable for high-resolution satellite images, where quadratic attention costs can become prohibitive.

4) *Differentiated Constraint for Robust Conditional Generation*: A major challenge in conditional diffusion-based cloud removal is conditional collapse, where the model tends to replicate the cloudy input due to high spatial similarity in noncloud regions. The proposed differentiated cloud-free feature loss mitigates this issue by discouraging excessive similarity between the generated output and the conditional input in uncertain regions. Rather than relying on explicit cloud masks during inference, this mechanism introduces a soft differentiation constraint during training, encouraging the model to reconstruct ambiguous areas actively while preserving consistency elsewhere. This improves training stability and prevents shortcut learning.

5) *Adaptive and Interpretable Fusion Via Sparse Gate*: The Sparsity-Aware Gated Fusion (SAGF) module is introduced to regulate how and where fused optical-SAR features are injected into the diffusion backbone network. Ablation experiments demonstrate that, despite lacking explicit cloud supervision, the learned gate mapping exhibits clear spatial selectivity.

The closer the gate value is to 1, the more the model tends to extract and utilize the conditional features fused by the MCMF module at that sampling point; conversely, it emphasizes retaining baseline features. In the shallow model (128×128 pixels²), the feature map preserves high-frequency texture information. Here, SAGF guides the model to fully extract texture and detail features from cloud-free regions of multcloud optical images while suppressing interference from cloud areas. At deeper levels (32×32 pixels²), semantic features become highly abstracted and local textures tend to blur. At this stage, SAGF prompts the network to shift its focus toward the global structural information provided by SAR beneath cloud cover. This demonstrates SAGF's ability to dynamically balance the fusion weights of multimodal conditional information within the network according to different scale-specific task requirements, with results highly consistent with empirical observations reported in ablation studies.

6) *Limitations and Future Work*: Despite its effectiveness, OS-MambaDiff inherits the computational overhead of iterative diffusion sampling, which limits the efficiency of inference. Future research will explore accelerated sampling strategies, such as step reduction and distillation, as well as extending the

framework to multitemporal and multisensor fusion scenarios. Broadening the application of the proposed design principles to other remote sensing restoration tasks also represents a promising direction.

V. CONCLUSION

This article proposes a novel solution named OS-MambaDiff for the problem of cloud contamination in optical remote sensing images. The model is based on the conditional diffusion framework and introduces Mamba to cloud removal for the first time, aiming to solve the challenges of dynamic multimodal information fusion and cloud contamination copy suppression encountered by current remote sensing cloud removal methods. The core contribution of this study is the design of an efficient Optical-SAR fusion module based on Mamba, which enhances the fusion capability of multimodal information while improving computational efficiency. Furthermore, we designed the SAGF mechanism, which encourages sparsity and decisive gating behavior. Without a cloud mask, the model is allowed to selectively determine the weight of conditional fusion information in different regions. We also propose the Differentiated Cloud-Free Feature Loss, to address the common issue of cloud contamination copying in conditional diffusion cloud removal models. This loss term forces the model to perform true feature reconstruction in cloud-covered areas, significantly improving the robustness and realism of the cloud removal results. Experimental results show that OS-MambaDiff surpasses existing Optical-SAR fusion cloud removal methods in all metrics, proving the effectiveness of the model. However, due to the complexity of geographical features in remote sensing images, the low computational efficiency of the diffusion model remains a problem. The inherent iterative denoising process leads to a slower inference speed. In the future, we will continue to research solutions that meet the requirements of real-time applications.

V. CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Peizhen Yang received the B.S. degree in remote sensing science and technology in 2018 from the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu, China, where he is currently working toward the Ph.D. degree in surveying and mapping science and technology.

His research interests include remote sensing image processing, multimodal image fusion, and multimodal image registration.



Bo Huang (Member, IEEE) received the B.Eng. degree in urban planning from Wuhan University, Wuhan, China, and the Ph.D. degree in remote sensing from the Chinese Academy of Sciences, Beijing, China.

He is currently a Chair Professor with the Department of Geography and the Deputy Director of the Urban Systems Institute, The University of Hong Kong, Hong Kong, SAR, China. His research interests include unified image fusion for sustainable urban environment, spatiotemporal data analytics, and multiobjective optimization for spatial planning with consideration of vulnerability and resilience.

Dr. Huang serves as an Asia-Pacific Editor for *International Journal of Geographical Information Science* (Taylor & Francis) and the Editor-in-Chief for *Comprehensive GIS* (Elsevier), a three-volume GIS sourcebook.

Dr. Huang serves as an Asia-Pacific Editor for *International Journal of Geographical Information Science* (Taylor & Francis) and the Editor-in-Chief for *Comprehensive GIS* (Elsevier), a three-volume GIS sourcebook.



Yuanxin Ye (Member, IEEE) received the B.S. degree in remote sensing science and technology from Southwest Jiaotong University, Chengdu, China, in 2008, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2013.

He is a Professor with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University. His research interests include remote sensing image processing, image registration, change detection, and object detection.

Dr. Ye was the recipient of the ISPRS Prizes for Best Papers by Young Authors of the 23rd International Society for Photogrammetry and Remote Sensing Congress, Prague, in 2016, and the Best Youth Oral Paper Award of ISPRS Geospatial Week 2017, Wuhan, in 2017. He is currently an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS).



Jiacheng Zhang received the B.S. degree in surveying and mapping science and technology in 2021 from Southwest Jiaotong University, Chengdu, China, where he is currently working toward the Ph.D. degree in surveying and mapping science and technology with the Faculty of Geosciences and Engineering.

His research interests include multimodal remote sensing image matching and fusion with related high-level vision tasks.



Min Zhao received the B.S. degree in electronic and information engineering and the M.S. and Ph.D. degrees in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2017, 2020, and 2024, respectively.

She is currently a Postdoctoral Fellow with the Department of Geography, The University of Hong Kong, Hong Kong. Her research interests include hyperspectral image analysis, image fusion, and object detection.

Dr. Zhao is the recipient of the People's Choice Award of Three Minute Thesis (3MT) Competition of IEEE IGARSS 2020, and the (with team) Champion of Grand Challenges on NIR Image Colorization of IEEE VCIP 2020.